

Test report #2

Environment description

OS: Windows 11
Interpreter: Python 3.13
Version: Git hash **c23eefcb** (16.06.2026)

System test scenarios

Scenario #0: Standard usage

The application works as expected when using as intended. The prompt leave no doubt how to use the application. Tables are clear, readable and properly formatted. CSV export works as expected and API calls successfully download the historical data.

Scenario #1: Not exact user inputs

ID	Test	User input	Expected result	Actual result
T01	Capitalization test	1 → usD, 1 → usd	Success	Success
T02	Whitespace test	1 → “USD “ → “ USD”	Success	Error

The application recognizes user input when its capitalization is different than specified, but does not when there's an extra whitespace character in the input („USD”, „USD ”, etc.).

Scenario #2: Invalid input types

ID	Test	User input	Expected result	Actual result
T03	Invalid currency	1 → ABC	Error	Error
T04	Out of range	0, 4, -1	Error	Error
T05	Invalid number type	0.5	Error	Error
T06	Invalid date		Error	Error
T07	Date in the future		Date too high	Insufficient date error
T08	Date too far in the past		Insufficient date error	No data found error

The application did not throw any unhandled errors when attempting to specify: Non-listed currencies, text in number inputs, floats in integer inputs, empty inputs in all fields, indices out of range and negative integers.

Date input from change distribution incorrectly identifies dates in the future as being too far in the past (For example: 31.12.2099). This is caused by the „start_date > min_date” check using the wrong sign also making the next check of „start_date > today” unreachable. As a consequence inputting a date too early returns a No data found error (**An error occurred: No data found for currency 'USD'. Check the currency code (e.g., USD, EUR, CHF).**)

Scenario #3: Repeated analysis

ID	Test	User input	Expected result	Actual result
T09	Repeated analysis	(1 → USD → 1 → n) x2	Success	Success
T10	Repeated save	(1 → USD → 1 → y) x2	Warning / Confirmation	File overwrite

When performing repeated analysis and exporting them no critical bugs were found. The only problem that might happen is that exporting results to a .csv file causes it to overwrite the previous one without asking or warning the user.

Scenario #4: Edge-case inputs

ID	Test	User input	Expected result	Actual result
T11	Self comparison	3 → USD → USD	Error	Error
T12	Choosing PLN after an error		Success	Error

The app correctly prevents the user from comparing the currency to itself even when the inputs differ in capitalization. When performing change distribution analysis and inputting an invalid currency, PLN disappears from the list of available currencies and cannot be chosen.

Scenario #5: No write permission

ID	Test	User input	Expected result	Actual result
T13	No file access	1 → USD → 1 → y	Handled error	Unhandled error

When the app has no write permissions and the user requests to download the table as .csv a raw error gets printed: An error occurred: [Errno 13] Permission denied: 'export_SESSION_ANALYSIS.csv'

Other Bugs

Bug #1 (Low Severity): Recursive input

ID	Test	User input	Expected result	Actual result
T14	Repetitive input error	"" x 10000	Nothing	Error

User reprompting for valid input happens recursively risking stack overflows.

Can be reproduced by holding enter in any of the input prompts.

```
INPUT:
INPUT INVALID (input y or n)
perform another analysis? (y/n)
INPUT: Traceback (most recent call last):
  File "C:\Users\Dadudze\PycharmProjects\ZPI2025_IO1_PoleHipNeed\main.py", line 5, in <module>
    app.run()
    ~~~~~^~
  File "C:\Users\Dadudze\PycharmProjects\ZPI2025_IO1_PoleHipNeed\data_processor.py", line 113, in run
    if not self.cli.ask_repeat():
    ~~~~~^~
  File "C:\Users\Dadudze\PycharmProjects\ZPI2025_IO1_PoleHipNeed\CLI.py", line 246, in ask_repeat
    return self.ask_repeat()
    ~~~~~^~
  File "C:\Users\Dadudze\PycharmProjects\ZPI2025_IO1_PoleHipNeed\CLI.py", line 246, in ask_repeat
    return self.ask_repeat()
    ~~~~~^~
  File "C:\Users\Dadudze\PycharmProjects\ZPI2025_IO1_PoleHipNeed\CLI.py", line 246, in ask_repeat
    return self.ask_repeat()
    ~~~~~^~
  [Previous line repeated 993 more times]
  File "C:\Users\Dadudze\PycharmProjects\ZPI2025_IO1_PoleHipNeed\CLI.py", line 237, in ask_repeat
    answer = self.get_input()
  File "C:\Users\Dadudze\PycharmProjects\ZPI2025_IO1_PoleHipNeed\CLI.py", line 127, in get_input
    return input()
RecursionError: maximum recursion depth exceeded
```

Retesting.

- The 0-division error in `calculate_change_distribution` has been fixed
 - The naming-convention bugs have not yet been fixed.
- https://github.com/IIS-ZPI/ZPI2025_IO1_PoleHipNeed/issues/73
- Redundant variable declarations have not yet been removed.
- https://github.com/IIS-ZPI/ZPI2025_IO1_PoleHipNeed/issues/73

Summary of found defects

- Inputs are not trimmed by whitespace.

https://github.com/IIS-ZPI/ZPI2025_IO1_PoleHipNeed/issues/68

- The start_date > min_date CLI.py:295 is misleading as it checks for the max date.

This check apparently misled the developers into adding wrong error messages, adding an always false check (start_date > today) and skipping the real start_date < min_date check.

https://github.com/IIS-ZPI/ZPI2025_IO1_PoleHipNeed/commit/0e77956fb3cdb36e4cf0f085a576daa54743553a

- Lack of overwrite warnings / confirmations.

https://github.com/IIS-ZPI/ZPI2025_IO1_PoleHipNeed/issues/70

- Improperly handled csv write errors.

https://github.com/IIS-ZPI/ZPI2025_IO1_PoleHipNeed/issues/60

- Input errors are handled recursively.

https://github.com/IIS-ZPI/ZPI2025_IO1_PoleHipNeed/issues/72

- PLN disappears from the list after an invalid input

https://github.com/IIS-ZPI/ZPI2025_IO1_PoleHipNeed/issues/71